Retail Sales Data Cleaning & Visualization

Complete Data Analysis Project

1. Project Objective

The objective of this project is to clean a raw retail sales dataset, handle missing values and duplicate records, detect outliers, process the data, and create visualizations to understand sales performance and customer behavior.

2. Technologies Used

- Python
- Pandas – data cleaning and analysis
- NumPy – numerical operations
- Matplotlib – data visualization
- Seaborn – advanced visualization

3. Project Workflow

Raw Dataset → Data Cleaning → Data Processing → Outlier Detection → Data Analysis → Visualization → Insights

4. Dataset Description

Column	Description
Order_ID	Unique order number
Date	Order date
Product	Product purchased
Category	Product category
Quantity	Number of items
Price	Price per item
Sales	Total sales
Region	Sales region
Customer_Rating	Customer rating

5. Data Cleaning

- Load the raw CSV dataset.
- Check for missing values.
- Fill missing numerical values using the median.

- Fill missing categorical values using the mode.
- Remove duplicate records.
- Convert the Date column into datetime format.

6. Outlier Detection

The project identifies unusual observations in Quantity, Price, Sales, and Customer_Rating. The Interquartile Range (IQR) method can be used to detect these outliers.

```
IQR = Q3 − Q1
Lower Bound = Q1 − 1.5 × IQR
Upper Bound = Q3 + 1.5 × IQR
```

7. Data Processing

Additional columns are created to make analysis easier:

```
Total_Sales = Quantity × Price
Month = Month extracted from Date
```

8. Data Visualization

- Monthly sales trend
- Sales by category
- Sales by region
- Top 10 products by sales
- Customer-rating distribution
- Quantity distribution
- Price vs Sales relationship

9. Analysis Questions

- Which category generates the highest sales?
- Which region performs best?
- Which products are most popular?
- Which month has the highest sales?
- Are there unusual or outlier transactions?
- Is there any relationship between price and sales?
- What is the average customer rating?

10. Expected Insights

After completing the analysis, the visualizations and summary statistics should reveal the strongest product categories, best-performing regions, high-value products, monthly sales patterns, unusual transactions, and customer-rating patterns.

11. Project Folder Structure

```
Retail-Sales-Data-Analysis/
■
■■■ dataset/
■   ■■■ retail_sales_raw.csv
■
■■■ output/
■   ■■■ cleaned_retail_sales.csv
■   ■■■ sales_visualizations/
■
■■■ retail_sales_analysis.py
■
■■■ README.md
```

12. Resume Project Description

Developed a Retail Sales Data Analysis project using Python, Pandas, NumPy, Matplotlib, and Seaborn. Cleaned raw data by handling missing values, duplicates, and outliers, performed data preprocessing, and generated visualizations to identify sales trends, top-performing products, categories, and regions.

13. LinkedIn Project Title

Retail Sales Data Cleaning, Analysis & Visualization using Python